

Solutions – Quiz 3

ECON 308 – International Economic Relations

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Multiple Choice (1 point each)

1. **(c) Factor endowments**

Differences in trade patterns arise because countries differ in relative factor supplies, not in technology or preferences.

2. **(b) Canada exports capital-intensive goods**

The Heckscher–Ohlin theorem predicts a capital-abundant country exports the capital-intensive good.

3. **(c) Trade raises the real income of the abundant factor and lowers that of the scarce factor**

According to Stolper–Samuelson, trade benefits the abundant factor and harms the scarce one.

4. **(b) Increase output of the labor-intensive good and decrease output of the capital-intensive good**

Rybczynski theorem: increasing a factor increases output of the good intensive in that factor, holding prices fixed.

5. **(b) The U.S. imported labor-intensive goods despite being capital-abundant**

The Leontief Paradox contradicted the simple H–O prediction for U.S. trade.

Problem (15 points)

Given:

- Home: $K_H = 300$, $L_H = 300$
- Foreign: $K_F = 100$, $L_F = 300$

Unit Input Requirements:

Good	Capital per unit (a_K)	Labor per unit (a_L)
Computers (C)	6	3
Textiles (T)	2	4

(a) Determine which country is capital- and labor-abundant (3 pts)

$$\frac{K_H}{L_H} = \frac{300}{300} = 1, \quad \frac{K_F}{L_F} = \frac{100}{300} = 0.33$$

Since $\frac{K_H}{L_H} > \frac{K_F}{L_F}$, Home has a relatively larger capital supply.

$\Rightarrow$ Home is capital-abundant, Foreign is labor-abundant.

(b) Identify which good is capital-intensive (3 pts)

Compute capital-labor ratios by sector:

$$\left(\frac{a_K}{a_L}\right)_C = \frac{6}{3} = 2, \quad \left(\frac{a_K}{a_L}\right)_T = \frac{2}{4} = 0.5$$

Since $2 > 0.5$, Computers use more capital per worker.

$\Rightarrow$ Computers are capital-intensive, Textiles are labor-intensive.

(c) Predict the trade pattern using Heckscher–Ohlin theorem (4 pts)

The Heckscher–Ohlin theorem states that each country exports the good that is intensive in its abundant factor. Therefore:

Home: Exports Computers (C), Imports Textiles (T)

Foreign: Exports Textiles (T), Imports Computers (C)

(d) Apply the Stolper–Samuelson theorem to Home (5 pts)

Step 1: Trade raises the relative price of Home’s export, Computers:

$$\frac{P_C}{P_T} \uparrow$$

Step 2: Under competitive pricing:

$$P_C = c_C(w, r), \quad P_T = c_T(w, r)$$

where w and r are the real wage and return to capital.

Step 3: Stolper–Samuelson states:

$$\frac{P_C}{P_T} \uparrow \Rightarrow \frac{r}{w} \uparrow$$

Thus, the real return to capital rises and the real return to labor falls.

Step 4: Economic interpretation:

- Capital owners (abundant factor) gain from trade.
- Workers (scarce factor) lose in real terms.

$$\Rightarrow \text{In Home: } r \uparrow, w \downarrow$$